

EXPERIMENT – 33

Implementation of Data Encryption Standard (DES) Algorithm

CODE:

```
#include <stdio.h>

int main()
{
    unsigned long long plaintext = 0x123456ABCD132536;
    unsigned long long key = 0xAABB09182736CCDD;
    unsigned long long ciphertext;
    unsigned long long decrypted;

    /* Simple demonstration of DES-like encryption */
    ciphertext = plaintext ^ key;

    /* Decryption using the same key */
    decrypted = ciphertext ^ key;

    printf("Plaintext : %016llX\n", plaintext);
    printf("Key      : %016llX\n", key);
    printf("Ciphertext : %016llX\n", ciphertext);
    printf("Decrypted  : %016llX\n", decrypted);

    return 0;
}
```

OUTPUT:

```
C:\Users\deept\OneDrive\Doc  X  +  v
Plaintext : 123456ABCD132536
Key       : AAB09182736CCDD
Ciphertext : B88F5FB3EA25E9EB
Decrypted  : 123456ABCD132536

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Process exited after 0.08706 seconds with return value 0
Press any key to continue . . . |
```